Kartik Jain

Software Engineer | Backend & Security

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Summary

Software engineer who builds and secures production systems: shipped a white-label learning platform as two Google Play apps, maintains a college system for ~4,000 students, and earned a $1,500 bug bounty from Meta.

Education & Leadership

B.Tech, Information Technology | *SVKM's NMIMS MPSTME, Mumbai* | **CGPA: 8.34/10.0** Expected Apr 2028

- **Chairperson**, ACM MPSTME (~250 members); ex-Technical Head: Taqneeq, Mumbai MUN, Social Conclave, Sattva.
- Built ACM's Coolify + Mailcow servers: @mpstmeacm.com mail for all members and 7+ committees.

Experience

VisionRise Technosoft Pvt. Ltd. | *Software Engineer Intern* Apr 2026 – Oct 2026

- Built a white-label learning platform end to end: a Node.js/Express/TypeScript REST API (276 routes, 20 modules, PostgreSQL + Drizzle ORM) and a React Native/Expo app shipped as two Google Play apps (Rise Academy, eGuru).
- Secured paid video with AES-128-encrypted HLS (FFmpeg), per-session playback tokens and signed Google Cloud Storage URLs; built JWT authentication with refresh tokens, Google Sign-In, role-based access control and audit logs.
- Provisioned and hardened the production Linux server (Ubuntu 24.04 LTS: key-only SSH, firewall, auto security updates); deployed Docker apps and PostgreSQL via Coolify CI/CD with automated backups to GCS.
- Integrated Zoom live classes (signature-verified webhooks, attendance, recordings) and Gemini quiz generation; administered the GitHub org (15 repos) and Google Workspace (35+ accounts).

R.U.D.R.A Cybersecurity Pvt. Ltd. | *Security Engineer Intern* May 2025 – Jul 2025

- Found 2 high-severity vulnerabilities (Local File Inclusion, broken access control) in a banking client's web portal through manual code review, then wrote custom Nuclei templates to detect them automatically.
- Automated external attack-surface monitoring with BBOT, saving ~$20,000/year of manual reconnaissance.
- Developed a custom Semgrep (SAST) module with Neo4j graph visualization for large-scale code analysis; ran purple-team exercises on the Sliver C2 framework.

Freelance | *Backend & Security Engineer* May 2025 – Oct 2026

- Led ransomware incident response for a 16,000+ user platform: restored it from backups and hardened its servers.
- Delivered a live player-auction and scoring platform (Next.js, Supabase Postgres + Realtime) used at ELEV8, a ~200-player, 8-team pickleball tournament, with an offline write queue for unreliable venue Wi-Fi.

Projects

GhostShare | end-to-end-encrypted file sharing | *TypeScript, Next.js, Fastify, PostgreSQL, Redis, Docker* Live

- Zero-knowledge design: keys are derived in-browser (Argon2id) and data is sealed with AES-256-GCM, so the server stores only ciphertext; encrypted 5 MiB chunks upload in parallel to Cloudflare R2 and resume after failures.

Railway Concession System | MPSTME's official portal | *Next.js, PostgreSQL, Drizzle* Live

- Took over and maintain the system ~4,000 students use: migrated it to PostgreSQL, enforced an IST-aware booking window server-side, and automated daily collection reports for the college office.

Event & Payment Backends | NMIMS student events | *Python, FastAPI, Flask, PostgreSQL* Delego API | Payments

- Lead backend author of Mumbai MUN's Delego app (600+ delegates, 2024–25) and Taqneeq's credits engine (OTP auth, referrals, cached leaderboard); co-led paid registration with HMAC-verified Cashfree payment webhooks.

bad-db | storage engine from scratch: custom file format, B-Tree index, compaction, own query language | *Java* Source

Security Research & Awards

- Awarded a **$1,500 bug bounty by Meta** for a privilege-escalation vulnerability in WhatsApp.
- Reported access-control and information-disclosure flaws in Microsoft Intune and Exchange Admin to MSRC; responsibly disclosed a stored XSS on the Mumbai Police website through CERT-In.
- **Won 3 hackathons:** MAL 4 (Paytm, team of 3), chat.win AI Jailbreakathon (global), CodeForge 2026 (team).

Skills

Core: Python, TypeScript, JavaScript, Java, Go, System Design (HLD/LLD, ADRs), Data Structures & Algorithms
Backend & Mobile: Node.js, Express, FastAPI, Flask, Next.js, React, React Native (Expo), REST APIs, JWT/OAuth 2.0
Data & Cloud: SQL, PostgreSQL, Redis, MongoDB, Drizzle ORM, Firebase, Google Cloud Storage, Cloudflare R2
DevOps: Docker, Linux, Bash, Git, GitHub Actions, CI/CD, Unit Testing, Coolify, Nginx, Backups & Disaster Recovery
Security: Web Application Security, Penetration Testing, VAPT, OWASP Top 10, Burp Suite, Nmap, SAST/DAST (Semgrep, Nuclei), SCA, Secure Code Review, Incident Response, Responsible Disclosure, SPF/DKIM/DMARC